

Chankyu (Charlie) Han

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RESEARCH INTERESTS

Smart Sensing and Haptic with AI/ML, Tactile Sensing, Human-Computer Interaction, Fabrication

EDUCATION

University of Washington (UW)

Seattle, Washington

Ph.D. in Electrical and Computer Engineering, Advisor: Dr. Yiyue Luo

Jun.2025 - Present

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, Korea

M.S. in Mechanical Engineering, Advisor: Dr. Inkyu Park

Mar.2021 - Feb.2023

B.S. in Mechanical Engineering

Sep.2016 - Feb.2021

Graduation with *Cum Laude*

PUBLICATIONS

([†]: equal contribution)

[13] [MagBall: Magnetic Rollerball for Multi-Scale Contact Interactions on Diverse Surfaces](#)

C. Han, Y. Miao, S. Zhang, Y. Luo

CHI 2026: ACM Conference on Human Factors in Computing Systems. <https://doi.org/10.1145/3772318.3791366>

[12] [Modular Tactile Sensing Platform with Sequential Optical Sampling for Human-Robot Interfaces](#)

H. Kim[†], S. Cho[†], **C. Han**, J. Ahn, D. Lee, S. Jang, G. Lee, K. Lee, H. Han, J.-H. Ha, J. Ahn, Y. Jeong, J. Choi^{*}, I. Park^{*}

Devices (2026). <https://doi.org/10.1016/j.device.2026.101081>

[11] [Scalable In-situ Fabrication of Multimodal Electronic Skin for Intelligent Robotics and Interactive System](#)

H. Lim[†], J. Choi[†], **C. Han**[†], D. Kim, H. Jin, M. Kim, Y. Kim, J. Yang, S. Seo, J. Jung, H. Ju, C. Hong, D. Lee, J. Ahn^{*}, H. Kim^{*}

npj Flexible Electronics (2026). <https://doi.org/10.1038/s41528-026-00538-4>

[10] [Thermoforming 2D Films into 3D Electronics for Customizable Tactile Sensing](#)

J. Choi[†], **C. Han**[†], D. Lee[†], H. Kim, G. Lee, J.-H. Ha, Y. Jeong, J. Ahn, H. Park, H. Han, S. Cho, J. Gu, I. Park

Science Advances 11.20 (2025): eadv0057. <https://doi.org/10.1126/sciadv.adv0057>

[9] [Spike-based Self-calibration for Enhanced Accuracy in Self-powered Pressure Sensing](#)

C. Han[†], J. Choi[†], J. Ahn, H. Kim, J.H. Ha, H. Han, S. Cho, Y. Jeong, J. Gu, I. Park

Advanced Materials Technologies (2023): 2301199. <https://doi.org/10.1002/admt.202301199>

[8] [Recent Advances in Sensor-Actuator Hybrid Soft Systems: Core Advantages, Intelligent Applications, and Future Perspectives](#)

C. Han[†], Y. Jeong[†], J. Ahn[†], T. Kim, J. Choi, J.-H. Ha, H. Kim, S.H. Hwang, S. Jeon, J. Ahn, J.T. Hong, J.J. Kim, J.H. Jeong, I. Park

Advanced Science (2023): 2302775. <https://doi.org/10.1002/advs.202302775>

- [7] **Nanotransfer Printing of Functional Nanomaterials on Electrospun Fibers for Wearable Healthcare Applications**
J.-H. Ha, J. Ko, J. Ahn, Y. Jeong, J. Ahn, S. Hwang, S. Jeon, D. Kim, S. A. Park, J. Gu, J. Choi, H. Han, **C. Han**, B. Kang, B.-H. Kang, S. Cho, Y.J. Kwon, C. Kim, S. Choi, G.-D. Sim, J.-H. Jeong, I. Park
Advanced Functional Materials (2024): 2401404. <https://doi.org/10.1002/adfm.202401404>
- [6] **Wireless, Multimodal Sensors for Continuous Measurement of Pressure, Temperature, and Hydration of Patients in Wheelchair**
S. Cho, H. Han, H. Park, S.-U. Lee, J.-H. Kim, S.W. Jeon, M. Wang, R. Avila, Z. Xie, K. Ko, M. Park, J. Lee, M. Choi, J.-S. Lee, W.G. Min, B.-J. Lee, S. Lee, J. Choi, J. Gu, J. Park, M.S. Kim, J. Ahn, O. Gul, **C. Han**, K. Lee, S. Kim, K. Kim, J. Kim, C.-M. Kang, J. Koo, S.S. Kwak, S. Kim, D.Y. Choi, S. Jeon, H.J. Sung, Y.B. Park, Y.T. Choi, M. Je, Y.S. Oh, I. Park
npj Flexible Electronics 7.1 (2023): 8. <https://doi.org/10.1038/s41528-023-00238-3>
- [5] **Battery-Free, Wireless, Ionic Liquid Sensor Arrays to Monitor Pressure and Temperature of Patients in Bed and Wheelchair**
H. Han, Y.S. Oh, S. Cho, H. Park, S.-U. Lee, K. Ko, J.-M. Park, J. Choi, J.-H. Ha, **C. Han**, Z. Zhao, Z. Liu, Z. Xie, J.-S. Lee, W.G. Min, B.-J. Lee, J. Koo, D.Y. Choi, M. Je, J.-Y. Sun, I. Park
Small 19.9 (2023): 2205048. <https://doi.org/10.1002/smll.202205048>
- [4] **All-Recyclable Triboelectric Nanogenerator for Sustainable Ocean Monitoring Systems**
J. Ahn[†], J.-S. Kim[†], Y. Jeong, S. Hwang, H. Yoo, Y. Jeong, J. Gu, M. Mahato, J. Ko, S. Jeong, J.-H. Ha, H.-S. Seo, J. Choi, M. Kang, **C. Han**, Y. Cho, C. H. Lee, J.-H. Jeong, I.-K. Oh, I. Park
Advanced Energy Materials (2022): 2201341. <https://doi.org/10.1002/aenm.202201341>
- [3] **A Review of Recent Advances in Electrically-Driven Polymer-based Flexible Actuators: Smart Materials, Structures, and Their Application**
J. Ahn[†], J. Gu[†], J. Choi[†], **C. Han**, Y. Jeong, J. Park, S. Cho, Y. Oh, J.-H. Jeong, M. Amjadi, I. Park
Advanced Materials Technologies (2022): 2200041. <https://doi.org/10.1002/admt.202200041>
- [2] **Stretchable Printed Circuit Board Based on Leak-Free Liquid Metal Interconnection and Local Strain Control**
M. S. Kim, S. Kim, J. Choi, S. Kim, **C. Han**, Y. Lee, Y. Jung, J. Park, S. Oh, B.-S. Bae, H. Lim, I. Park
ACS Applied Materials & Interfaces 14.1 (2021): 1826-1837. <https://doi.org/10.1021/acsami.1c16177>
- [1] **Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming**
J. Choi, **C. Han**, S. Cho, K. Kim, J. Ahn, D. D. Orbe, I. Cho, Z. Zhao, Y. Oh, H. Hong, S. S. Kim, I. Park
Science Advances 7.42 (2021): eabj0694. <https://doi.org/10.1126/sciadv.abj0694>

PATENT & TECHNOLOGY TRANSFER

Joint transfer of the following technologies (\$37,000, 2022):

Finger wearable touch sensor, control system using same, and method thereof

I. Park[†], J. Choi[†], **C. Han**[†], G. Lee[†]

Korea Patent (Application Number: 10-2021-0096964), 2021.

Large-Area Multi-Pressure Sensor Pad Fabrication Technology (Know-how)

I. Park[†], J. Choi[†], **C. Han**[†], G. Lee[†]

RESEARCH EXPERIENCES

UW Wearable Intelligence Lab Research Assistant Mentor: Yiyue Luo	Jun.2025 - Present
Korea University Sensor and Nanofabrication System Lab Visiting Researcher Mentors: Junseong Ahn, Junrak Choi	Jan.2025 – Jun.2025
KAIST Machine Intelligence and Novel-sensor Technology Lab Undergraduate Intern & Research Assistant Mentor: Inkyu Park	Jun.2020 – Jun.2023

SCHOLARSHIPS & AWARDS

Best Oral Presentation Award IEEE NEMS 2023 [9]	May 2023
Best Poster Award International Conference on Advanced Electromaterials (ICAE) 2021 [1]	Nov. 2021
National Excellence Scholarship (Graduate Program) Government funding for admission fees, full amount of school support fees in the master's program	Mar.2021 – Feb.2023
KAIST Undergraduate Research Grant Title: Customizable, Conformal, and Stretchable 3D Electronics via Pre-distorted Pattern Generation and Thermoforming [1]	Sept. 2020 - Feb. 2021
Dean's List, School of Freshman Top 2% of KAIST freshman cohort	Feb. 2017
National Excellence Scholarship (Undergraduate Program) Government funding for admission fees, full amount of school support fees in the undergraduate program	Sep.2016 – Feb.2020

SERVICE

Reviewer Nature Communications (1), CHI 2026	
Teaching Assistant UW EE 241 Programming for Signal and Information Processing Applications KAIST ME 340 Engineering Design	
Outreach and Leadership CHI Workshop Organizer: Everyday Wearable for Personalized Health and Well-Being High School Outreach: Research Talk and Lab Tour on Wearable Intelligence	2026 2025

MEDIA COVERAGE

Bridge Economy (Viva100) “Multimodal Electronic Skin” [11]	Feb. 2026
KAIST ME News “Tactile Sensor Detecting Light Breath, Pressure, and Sound” [10]	Jul. 2025

SKILLS

Languages: Python, C/C++, MATLAB, LabVIEW

Fabrication: 3D printing (DLP, FDM), Thermoforming, Laser processing(CO₂, UV, Fiber), PCB design

TOEFL: 108/120

OTHERS

Military Service in Republic of Korea Army	Jun.2023 – Dec.2024
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